FFRViT: Frequency Feature Refinement Vision Transformer for Remote Sensing Object Detection

Lihua Fu, Wenwen Liu, Guofang Li, and Weijin Huang

***Abstract*—Rotated object detection in remote sensing images is a research hot spot in the field of computer vision. The following problems exist in the detection of rotating object in remote sensing images: insufficient feature extraction ability, complex scenes lead to interference and occlusion between objects, and object characteristics are sensitive to positioning. To solve the above problems, we propose the frequency feature refinement vision transformer (FFRViT). First, the initial feature extraction is performed on the input image. Then, the frequency division feature refinement of local, global, and contextual features is performed in four stages. This process not only emphasizes the key positional information of the image but also models global relationships and extracts surrounding environmental information as a supplement, thereby enhancing the model's ability to capture comprehensive features. Finally, the detection head is used for object classification and localization. A sample collaborative optimization loss is designed to adaptively adjust the model's emphasis on positive and negative samples, so as to further improve the detection accuracy of the model. Experiments on the HRSC2016 and DOTA-v1.0 public datasets show that the proposed method achieves better performance in rotated object detection in remote sensing images, and its detection accuracy is better than some mainstream methods.**

***Index Terms*—Feature interaction, feature refinement, rotated object detection, vision transformer.**

I. Introduction

ROTATE object detection is one of the fundamental tasks in remote sensing image processing, aimed at identifying the categories of one or more specific objects in a given remote sensing image and locating them accurately [1], [2], [3]. Compared with general object detection, rotated object detection can provide more precise information regarding the orientation and scale of the targets. It plays a critical role in fields, such as intelligence reconnaissance, disaster monitoring, urban planning, and so on, while also serving as an essential foundation for tasks, such as object tracking and image segmentation.

In recent years, the rotated object detection techniques based on deep learning are mainly divided into two categories [4]: convolution-based rotated object detection and transformer-based rotated object detection. Convolution-based methods for rotated object detection often use predefined anchor points and prior boxes or directly forecast keypoints, with convolutional modules extracting features and successfully capturing spatial local information in the image. On the other hand, transformer-based methods introduce self-attention mechanisms to enhance feature representation capability, expand the receptive field of the image, and model global relationships, enabling the integration of long-range dependencies.

Despite the significant progress made in rotated object detection for remote sensing images, there are still several challenges that need to be addressed.

1) The pure convolution method can only capture local features within an image, while transformer is capable of establishing long-range dependencies across the entire image. However, transformer lacks translation invariance and the ability to extract local information, which will affect the detection accuracy.
2) Remote sensing image scene is complex with diverse target scales and uneven distributions. Vehicles, ships, and other objects in the image are often densely arranged, which leads to mutual interference between the objects, which makes the accurate detection of rotating objects more complicated.
3) The objects of remote sensing images include bridges, ships, ports, and so on. The aspect ratio of these objects is too large, and the slight angle deviations can cause a rapid decrease in their intersection over union (IoU) value. Additionally, the classification and regression are typically trained independently, which has the problem of classification-regression inconsistency, that is, classification scores cannot directly reflect the quality of the predicted bounding boxes.

To address the above problems, the main contributions of this article are as follows.

1) To achieve high-precision detection of dense rotated objects, we design a four-stage frequency feature refinement process. Based on the characteristics of convolution and transformer, the features of high-frequency, low-frequency, and super-pixel components in remote sensing images are extracted. This effectively integrates the local, global, and contextual information of the targets, enhancing the model's detection accuracy and robustness in complex scenes.
2) To align the model's classification and regression scores, we design positive/negative sample weight

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factors and propose sample collaborative optimization loss to adaptively optimize the model's learning weight between positive and negative samples, better balance the classification and regression branches, and alleviate inconsistency of classification regression in object detection.

3) The proposed method achieves advanced performance on both HRSC2016 and DOTA-v1.0 public benchmark datasets, demonstrating superior performance compared with other rotated object detection methods.

II. Related Work

Rotated object detection can be categorized into convolution-based and transformer-based approaches. The convolution-based methods are further divided into anchor-based and anchor-free methods.

Anchor-based rotated object detection assigns anchor boxes with different aspect ratios at each feature point, which are then filtered and adjusted to obtain the final predicted bounding boxes. This approach achieves high detection accuracy but tends to be slower. ReDet [5] leverages a rotation-equivariant network and RiRoI alignment to extract rotation-invariant features in both spatial and directional dimensions. R3Det [6] designs a feature refinement module and pixel-level feature interpolation to further enhance detection performance. Oriented R-CNN [7] employs a lightweight module to generate oriented proposals and midpoint offset representations with competitive accuracy compared to advanced two-stage detectors, achieving the approximate efficiency of single-stage detectors. LSKNet [8] dynamically adjusts the spatial receptive field to better simulate the measurement environment of various objects in remote sensing scenes.

Anchor-free rotated object detection predicts key points to obtain the final predicted bounding boxes. These methods have fewer network parameters and faster detection speeds but tend to sacrifice some accuracy. Oriented RepPoints [9] designs adaptive points assessment and assignment to measure the quality of oriented points, considering classification, localization, orientation, and pointwise correlation, significantly improving prediction performance. PP-YOLOE-R [10] introduces rotation task alignment learning and decouples angle prediction heads to better accommodate rotated object detection. RTMDet [11] employs large convolutional kernels and a dynamic label assignment strategy with soft labels to enhance detection accuracy.

At present, many detectors based on convolutional backbone networks have achieved promising performance in the field of remote sensing object detection. PKINet [12] extracts multiscale texture features across different receptive fields using nondilated deep convolutions and incorporates a contextual anchoring attention mechanism to capture long-range contextual information. Strip R-CNN [13] employs orthogonal large strip convolutions to capture spatial information, thereby enhancing the localization capability of remote sensing object detectors. Although convolutional neural networks (CNNs) still dominate the field, transformer-based detectors have gradually emerged in recent years, bringing new research focus due to their powerful global modeling capability. Transformer networks have performed exceptionally well in natural language processing (NLP) tasks, significantly improving model processing capabilities [14]. Vision transformer (ViT) is a pioneering work that applies the transformer architecture to vision tasks, achieving the state-of-the-art results [15]. Carion et al. [16] proposed the DETR model, which was the first to apply the transformer architecture to object detection. AO2-DETR [17] introduced an oriented proposal generation mechanism, a refinement module, and a rotation-aware set matching loss. ARS-DETR [18] designed an aspect ratio-aware circular smooth label that dynamically computes the smoothing value based on the target's aspect ratio and proposed a rotation-adaptive deformable attention module, enabling sampling points to rotate accordingly and eliminating misalignment between region features and sampling locations. Oriented-DETR [19] extended the end-to-end transformer detection framework with point-axis prediction, where the point-axis representation effectively decouples position and orientation, providing a more flexible and continuous modeling approach. Beyond DETR-based models, STD [20] employs a separate network branch to predict the bounding box's location, size, and angle, while utilizing cascaded activation masks to progressively enhance features within the region of interest.

III. Proposed Method

A. Architecture

We propose a rotated object detection model based on frequency feature refinement. As shown in Fig. 1, the detection model mainly consists of a four-stage feature refinement process, each with two modules: frequency splitter and feature interaction module (FIM). The FIM includes positional convolution block (PCB), self-focusing transformer block (STB), and contextual transformer block (CTB).

The proposed model first extracts initial features from an input image of size $H \times W \times 3$ using the Stem section, which consists of two 3×3 depthwise separable convolutions. Next, the extracted initial features undergo a four-stage feature refinement process, progressing from coarse to fine, producing local features Y_L, global features Y_G, and contextual features Y_C. These three types of features are then concatenated along the channel dimension and passed through a 1×1 convolution to obtain the detection features. Finally, the region of interest is selected, and classification and regression are performed using fully connected and convolutional layers, respectively, to obtain the final object category and precise location, completing the detection task.

B. Frequency Splitter

CNN is the most basic structure for feature extraction. By applying local convolution in the receptive field, CNN can effectively capture high-frequency information [21]. Existing studies show that transformer tends to capture low-frequency information, mainly global shapes and structures, while they often fall short in extracting high-frequency information, such as local edges and texture details [22].

Therefore, in order to use CNN and transformer structures to extract features more effectively, we design the frequency

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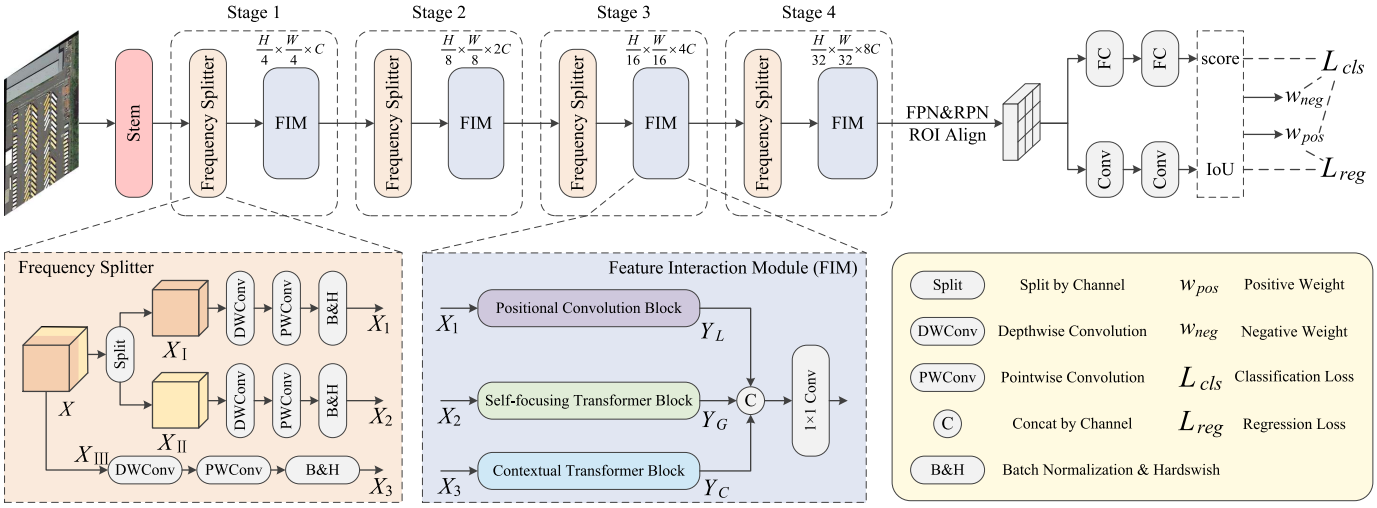


Fig. 1. Architecture of FFRViT.

splitter to extract corresponding features according to the characteristics of different structures. CNN is used to extract high-frequency features, while transformer is employed to capture low-frequency features. Since the acquisition of context relations between features is not purely global features and the range of receptive field is larger than common convolution, complete input features are used to extract context information.

First, the input feature $X \in \mathbb{R}^{H\times W\times C}$ is split along the channel dimension into high-frequency features $X_I \in \mathbb{R}^{H\times W\times (C/2)}$ and low-frequency features $X_{II} \in \mathbb{R}^{H\times W\times (C/2)}$, while the complete input feature $X_{III} \in \mathbb{R}^{H\times W\times C}$ is used as a third branch. Then, the features X_I, X_{II}, and X_{III} pass through depthwise convolution [23], pointwise convolution, batch normalization [24], and Hardswish [25] to obtain the output features X_1, X_2, and X_3, which are used as inputs for the PCB, STB, and CTB, respectively, in the FIM of this stage

$$X_i = H\left(\mathrm{BN}\left(PwC\left(DwC\left(X_j\right)\right)\right)\right),\quad i=1,2,3,\quad j=I,II,III \tag{1}$$

where $DwC(*)$ represents the depthwise convolution, $PwC(*)$ represents the pointwise convolution, BN$(*)$ represents the batch normalization layer, and $H(*)$ represents the Hardswish activation function.

C. Positional Convolution Block

Introducing attention mechanisms into the feature extraction process based on convolution can significantly enhance the model's performance. However, most attention mechanisms either lack long-range dependencies in the spatial dimension or reduce the channel dimension, making it difficult to effectively extract features [26], [27].

This article designs the PCB, which uses 1-D convolution to obtain attention maps for horizontal and vertical positions, thereby enhancing the local positional information in high-frequency features and improving the model's ability to capture spatial details. The structure of the PCB is shown in Fig. 2.

First, for the input high-frequency features X_1, average pooling is applied separately along the horizontal and vertical

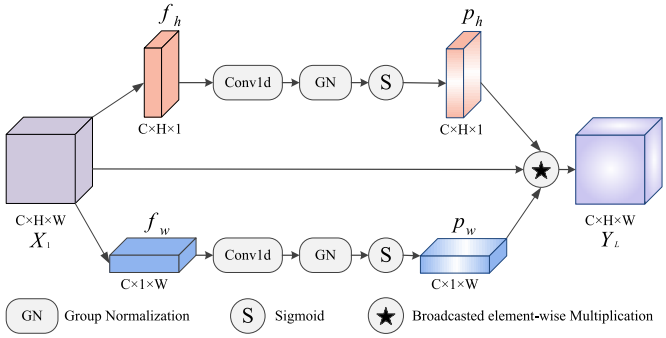


Fig. 2. Positional convolution block.

directions on each channel to obtain the features f_h and f_w

$$f_h(i) = \frac{1}{W}\sum_{0\le j<W} X_1(i,j) \tag{2}$$

$$f_w(j) = \frac{1}{H}\sum_{0\le i<H} X_1(i,j). \tag{3}$$

Then, 1-D convolution of feature f_h and f_w is used to enhance the position information in the horizontal and vertical directions, respectively, and then, group normalization [28] and the sigmoid function are used in turn to obtain the horizontal and vertical position attention graphs p_h and p_w

$$p_h = \sigma\left(\mathrm{GN}\left(\mathrm{Conv1d}\left(f_h\right)\right)\right) \tag{4}$$

$$p_h = \sigma\left(\mathrm{GN}\left(\mathrm{Conv1d}\left(f_w\right)\right)\right) \tag{5}$$

where σ represents the sigmoid activation function, GN represents the group normalization, and Conv1d represents the 1-D convolution with a kernel size of 7.

Since f_h and f_w are sequential signals in channels, 1-D convolution is more efficient and lighter than 2-D convolution. At the same time, clustering the input feature channels and then normalizing each group can improve the high dependence of batch size in batch normalization.

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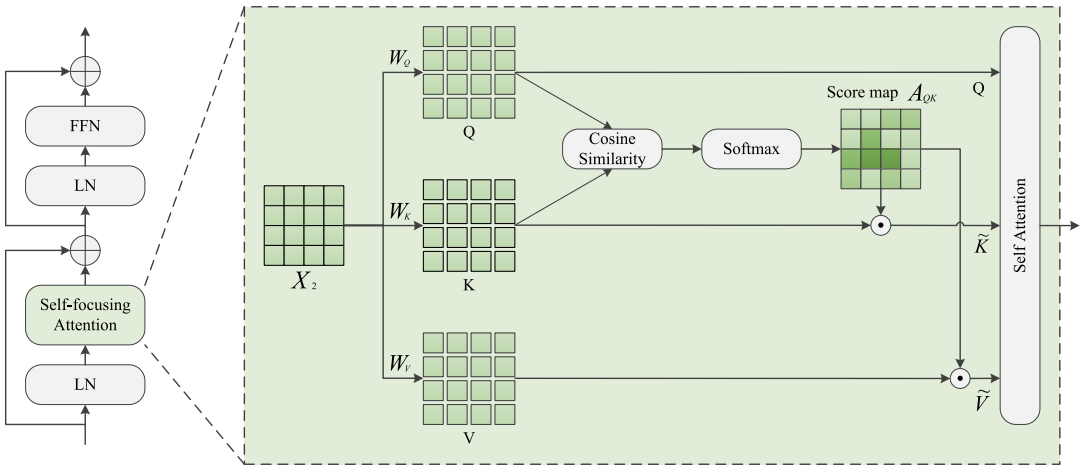


Fig. 3. Self-focusing transformer block.

Finally, the horizontal position attentional map p_h and vertical position attentional map p_w are taken as weights and multiplied with the input feature X_1 to improve the significant information of the horizontal and vertical directions, respectively, and the local feature Y_L is obtained

$$Y_L = X_1 * p_h * p_w \tag{6}$$

where X_1 is the input high-frequency feature, p_h is the horizontal position attention map, and p_w is the vertical position attention map.

D. Self-Focusing Transformer Block

When only local information is used as a detection feature in object detection, it lacks understanding of the relationship between the overall image and the target, and it is unable to deal with the complex scene effectively, which may result in missing, false detection, or inaccurate object detection. As a result, incorporating global information into object detection improves the model's accuracy in complicated scenes. Most existing studies use transformer to introduce a self-attention method for modeling global information.

Although the multihead attention mechanism in traditional transformer is capable of long-distance modeling, it ignores the internal connection between tokens, does not distinguish the importance or context of tokens in the sequence, and uses the same processing method for each token, rendering it incapable of effectively distinguishing the target region from the background region. This can result in the loss of information or the introduction of irrelevant noise information.

This article introduces a self-focusing attention mechanism and designs the STB to adaptively extract global features from the input low-frequency features. The structure of this block is illustrated in Fig. 3.

First, the input low-frequency feature X_2 is projected into query, key, and value representations

$$Q = X_2 W_Q \tag{7}$$

$$K = X_2 W_K \tag{8}$$

$$V = X_2 W_V. \tag{9}$$

Next, the cosine similarity of Q and K is calculated and normalized using Softmax function to obtain the significance score map A_{QK}

$$A_{QK} = \text{Softmax}\left(\frac{QK}{\|Q\|\,\|K\|}\right) \tag{10}$$

where $\|*\|$ represents the Euclidean norm of the feature.

Then, the significance score map A_{QK} is used to update K and V, resulting in the updated representations $\tilde{K}$ and $\tilde{V}$

$$\tilde{K} = KA_{QK} \tag{11}$$

$$\tilde{V} = VA_{QK}. \tag{12}$$

Finally, Q is calculated with the updated $\tilde{K}$ and $\tilde{V}$ for self-attention

$$\text{SGAttn}(X) = \text{Softmax}\left(\frac{Q\tilde{K}^T}{\sqrt{d_k}}\right)\tilde{V} \tag{13}$$

where d_k is the dimension of the matrix $\tilde{K}$ and is used as a scaling factor to prevent the dot product value from being too large and causing the gradient to disappear or explode.

STB uses the significance scores of query and key to update the values of key and value, respectively, and guide query to query in more relevant key–value pairs, thereby reducing the interference of redundant information.

E. Contextual Transformer Block

In remote sensing images, most of the objects are difficult to identify accurately based on their appearance only because of their small size. Existing research points out that limited background information often leads to misclassification. Therefore, the effective extraction of context information is particularly important to improve the performance of object detection.

In this article, the contextual attention mechanism is introduced and the CTB is designed to aggregate the pixel information of super-pixel features to extract contextual features. Its structure is shown in Fig. 4.

First, the feature $X_3 \in \mathbb{R}^{H\times W\times C}$ is passed through two separate branches for processing. One branch divides X_3 into windows of size d and then performs average pooling on

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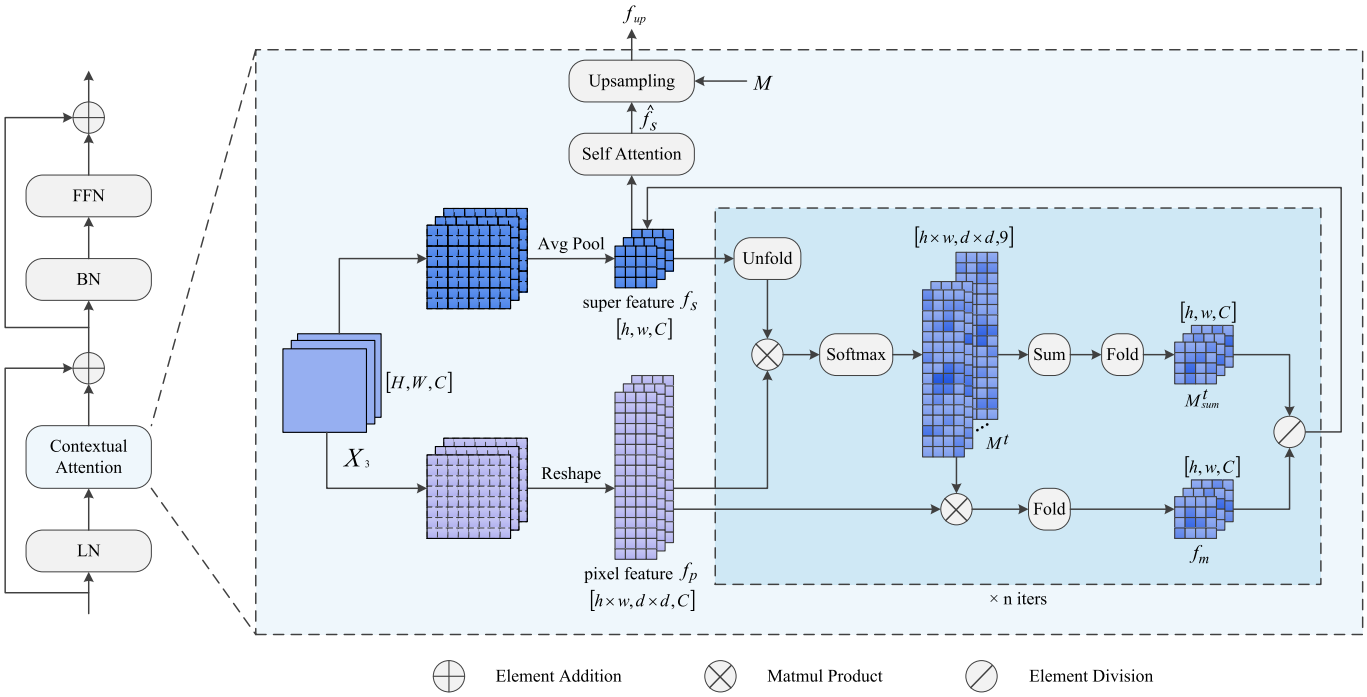


Fig. 4. Contextual transformer block.

each window to generate super-pixel features $f_s \in \mathbb{R}^{h\times w\times C}$. The other branch reconstructs X_3 to obtain pixel features $f_p \in \mathbb{R}^{(h\times w)\times(d\times d)\times C}$, where $h = (H/d)$ and $w = (W/d)$.

The core of the context attention mechanism lies in the iteration and update of super-pixel features. For example, the update process of the t iteration is as follows:

First, the super-pixel feature f_s^{t-1} after the $t-1$ iteration and update is expanded and reshaped into the appropriate shape to ensure that each token of the pixel feature f_p only needs to be calculated with its surrounding super-pixel token when calculating the affinity matrix M^t, thus reducing redundant operations. The affinity matrix M^t is calculated as follows:

$$f = \text{Unfold}\left(f_s^{t-1}\right) \in \mathbb{R}^{(h\times w)\times(C\times 9)} \tag{14}$$

$$M^t = \text{Softmax}\left(f_p \times \text{reshape}\left(f\right)\right) \tag{15}$$

where Unfold represents the use of convolution to extract the local receptive field around each super-pixel, and Softmax aims to obtain a normalized affinity distribution.

Then, calculate the sum of the affinity matrix, which is used to normalize the token, and reconstruct it back to its original shape by convolution

$$M^t_{\text{sum}} = \text{Fold}\left(\sum_{i=1}^{d\times d} M_i^t\right). \tag{16}$$

Finally, the affinity matrix M^t and pixel feature f_p are multiplied by matrix and divided by the appropriate elements of M^t_{sum} for normalization processing, yielding the updated super pixel feature f_s^t

$$f_s^t = \frac{f_m}{M^t_{\text{sum}} + \epsilon} \tag{17}$$

$$f_m = \text{Fold}\left(M^t \times f_p\right) \tag{18}$$

where ϵ is the numerical stability constant to avoid the numerical divisor error and reduce the floating-point error.

After n iterations and updates, the final super-pixel feature f_s is produced, and the self-attention computation is performed to obtain the feature $\widehat{f_s}$. Based on the affinity matrix M acquired in the last iteration, the feature $\widehat{f_s}$ is upsampled and mapped to the pixel token size

$$f_{\text{up}} = M \times \widehat{f_s} \tag{19}$$

$$\widehat{f_s} = \text{Softmax}\left(\frac{Q\left(f_s\right)K^T\left(f_s\right)}{\sqrt{d}}\right)V\left(f_s\right) \tag{20}$$

where d is the dimension of the matrix K, used as a scaling factor, and M represents the affinity matrix obtained from the last iteration.

F. Sample Collaborative Optimization Loss

Good prediction results require both high classification scores and precise location. However, due to the independent training of classification and regression branches, classification scores and positioning results are frequently misaligned, as illustrated in Fig. 5.

Therefore, if all training samples are treated equally, it may occur that the category of the prediction box has the highest score but the position is not the most accurate [29], [30]. In order to align the classification scores and regression scores, we introduce positive/negative sample weights to design sample collaborative optimization losses.

The positive sample weight ω_{pos} reflects the importance of positive sample in classification and localization. Obviously, ω_{pos} should be positively correlated with the classification score and IoU value, while using an exponential

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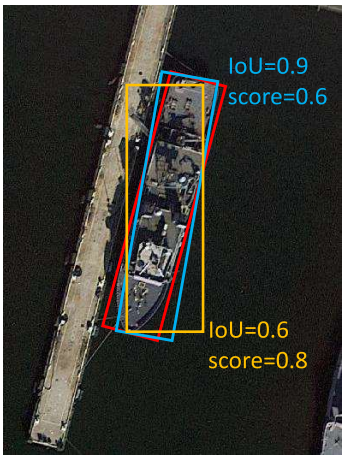


Fig. 5. Classification-regression inconsistency refers to situations where high classification scores do not correspond to accurate localization. For example, the blue box has an IoU of 0.9 and a confidence score of 0.6, while the yellow box has an IoU of 0.6 and a confidence score of 0.8. The yellow box has a higher classification score, but the blue box is more accurately localized (red box: ground-truth box; blue and yellow boxes: predicted boxes).

function to widen the weight difference. Therefore, ω_{pos} is set to

$$\omega_{\text{pos}} = e^{\alpha(s\times\tau)} \tag{21}$$

where s is the classification score, τ is the IoU value, and α is the hyperparameter.

The negative sample weight ω_{neg} consists of two parts: the probability and importance of the negative sample. When the IoU value of the anchor box is less than the threshold value, it is judged to be a negative sample. It can be seen that IoU is the only factor that determines the probability of the anchor box being a negative sample. Since the smaller the IoU, the more it is expected to be a negative sample, on the contrary, the closer the IoU value is to 1, the more it is expected to be a positive sample, so the monotone decreasing function is used to represent the probability that the anchor box is a negative sample. At the same time, in the inference stage, the prediction of negative samples will not affect the recall rate but will affect the accuracy, that is, the classification score of negative samples is as small as possible. Therefore, ω_{neg} is set to

$$\omega_{\text{neg}} = s \times \left(-\log(\tau)\right) \tag{22}$$

where s is the classification score and τ is the IoU value.

Therefore, the sample collaborative optimization loss $\mathcal{L}_{\text{det}}$ is the weighted sum of classification loss $\mathcal{L}_{\text{cls}}$ and regression loss $\mathcal{L}_{\text{reg}}$

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{cls}} + \lambda\mathcal{L}_{\text{reg}} \tag{23}$$

$$\mathcal{L}_{\text{cls}} = \sum_{n=1}^{N} -\omega_{\text{pos}}^{n} \times \log(s^{n}) - \omega_{\text{neg}}^{n} \times \log(1-s^{n}) \tag{24}$$

$$\mathcal{L}_{\text{reg}} = \sum_{n=1}^{N} \omega_{\text{pos}}^{n} \times \text{smooth}_{L_1}\left(b, b'\right) \tag{25}$$

where λ is the balance factor, N is the number of prediction boxes, s is the classification score, and b and b' are the positions of prediction boxes and ground-truth boxes, respectively.

IV. Experiments and Results

A. Datasets

In order to evaluate the validity of the proposed model, we conducted experiments on two widely used public benchmark datasets for remote sensing images: HRSC2016 [31] and DOTA-v1.0 [32].

1) HRSC2016: As a publicly available remote sensing image ship dataset, it is a rare remote sensing ship dataset that subdivides ship categories. This dataset contains 1061 effectively labeled images with a number of 2976 targets, with image sizes ranging from 300 × 300 to 1500 × 900, and 436, 181 and 444 images in the training, validation, and test sets, respectively.

2) DOTA-v1.0: This is a large-scale object detection dataset. For remote sensing images, it consists of 2806 aerial images, each with a pixel size between 800 × 800 and 4000 × 4000, containing 188282 instances of different scales, orientations, and shapes, with 15 categories marked in detail: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). The training set, validation set, and test set of this dataset contain 1411, 458, and 937 images, respectively.

B. Implementation Details

The experimental environment of this article is NVIDIA GeForce RTX 3090 GPU with a batch size of 2, and the model is built based on MMDetection, an open-source framework. In this article, the same dataset processing method as other mainstream methods was adopted. The backbone network is first pretrained on the ImageNet-1K dataset and then fine-tuned on the object remote sensing benchmark. AdamW optimizer was used to train 36 epochs on HRSC2016 data set and 12 epochs on DOTA-v1.0 data set, and the channel numbers in the four stages of the model were set to be 96, 192, 384, and 512, respectively.

When training on HRSC2016 and DOTA-v1.0, the initial learning rate of the model is 0.0004 and 0.0001, respectively, and the weight attenuation is 0.05. Meanwhile, a shape-adaptive selection (SAS) [33] like strategy is used to adaptively adjust the IoU threshold to divide positive and negative samples. All models are trained on the training and validation sets and tested on the test set, noting that the performance evaluation of the models on the DOTA-v1.0 dataset follows a multiscale setup. The ablation experiments of the model on the DOTA-v1.0 dataset are performed in a single-scale setting, and the model is trained on the training set and tested on the validation set. All images in the DOTA-v1.0 dataset are cropped into patches of size 1024 × 1024. For single-scale settings, a stride of 824 is used for cropping, while in multiscale settings, the original images are scaled to three levels (0.5, 1.0, 1.5) with a stride of 524.

C. Comparisons With the State-of-the-Art Methods

The proposed method is compared with several mainstream rotation object detection methods on two datasets to analyze detection performance.

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Fig. 6. Visualization of detection results on the HRSC2016 dataset.

TABLE I

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE HRSC2016 DATASET. 07 MEANS EVALUATED UNDER PASCAL VOC2007 METRIC, AND 12 MEANS EVALUATED UNDER PASCAL VOC2012 METRIC

Method	mAP(07)	mAP(12)
CenterMap [34]	-	92.70
RoI Trans. [35]	86.20	-
Gliding Vertex [36]	88.20	-
CSL [37]	89.62	96.10
R3Det [6]	89.26	96.01
DAL [38]	89.77	-
GWD [39]	89.85	97.37
S2ANet [40]	90.17	95.01
AOPG [41]	90.34	96.22
ReDet [5]	90.46	97.63
O-RCNN [7]	90.50	97.60
RTMDet-R [11]	**90.60**	97.10
Ours	90.59	**97.78**

1) Results on the HRSC2016 Dataset: On the HRSC2016 dataset, the detection accuracy of the proposed rotated object detection method is compared with other methods, as shown in Table I. The detection results of the comparison methods are obtained according to the original paper, in which red represents the highest precision and blue represents the second-highest precision.

From Table I, we can obtain the following results.

1) On the HRSC2016 dataset, the accuracy of mAP(07) of the proposed method reaches 90.59%, which is only 0.01% lower than the highest accuracy. The accuracy of the proposed method in mAP (12) is 97.78%, which is higher than that of other comparison methods and 0.15% higher than the second-highest accuracy.
2) The targets in the HRSC2016 dataset are mainly ships with arbitrary orientation and too large aspect ratio, which presents a challenge for rotated object detection. Experimental results show that the proposed method has high detection performance when dealing with targets with large aspect ratio, and its effectiveness in such object detection is verified.

The qualitative detection results of the proposed method on the HRSC2016 dataset are shown in Fig. 6. Since the dataset contains only one category of ships, all results are marked in a red box with the corresponding category and confidence score next to it.

From Fig. 6, we can obtain the following results.

1) For ships of different scale and distribution, the proposed method can detect well. It adopts a four-stage process from coarse to fine frequency feature refinement, which helps to gradually optimize the feature representation, so as to better capture the detailed information of the ship, and improve the recognition ability to various size and location targets. In addition, through the frequency feature interaction, not only the local information is retained, but also the global and contextual information is supplemented, which further improves the accuracy of model detection.
2) Ships with different aspect ratios were successfully detected, demonstrating the effectiveness of sample collaborative optimization loss based on positive/negative sample weights. By balancing classification and regression branches, classification-regression inconsistency in object detection was alleviated.

2) Results on the DOTA Dataset: Table II compares the detection accuracy of the proposed rotated object detection method to that of other methods using the DOTA-v1.0 dataset. The detection results of the comparative approaches are obtained based on the original paper. The proposed model contains 46.88 million parameters and incurs a computational cost of approximately 252.13 GFLOPs.

From Table II, we can obtain the following results.

1) The proposed method achieves the best performance in ship, storage tank, and HC, and the detection accuracy reaches 90.13%, 88.42%, and 79.79%, respectively. Compared with the best detection methods, the accuracy is increased by 0.88%, 0.04%, and 1.11%, respectively. In view of the fact that ship and HC are both targets with large aspect ratio, the sample collaborative optimization loss designed can retain more effective positive samples and promote the joint improvement of classification score and positioning accuracy.
2) The proposed method also achieves suboptimal performance in large vehicle and SP, with a detection accuracy of 86.06% and 79.68%, respectively, which is only 0.01% and 0.03% lower than the highest accuracy. In order to solve the problem of dense vehicle arrangement

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TABLE II
COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE DOTA DATASET

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
CenterMap [34]	89.83	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06	76.03
RoI Trans. [35]	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61
Gliding Vertex [36]	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
SCRDet [42]	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
CSL [37]	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93	76.17
R3Det [6]	89.80	83.77	48.11	66.77	78.76	83.27	87.84	90.82	85.38	85.51	65.67	62.68	67.53	78.56	72.67	76.47
S2ANet [40]	88.89	83.60	57.74	81.95	79.94	83.19	89.11	90.78	84.87	87.81	70.30	68.25	78.30	77.01	69.58	79.42
SASM [33]	89.54	85.94	57.73	78.41	79.78	84.19	89.25	90.87	58.80	87.27	63.82	67.81	78.67	79.35	69.37	79.17
ReDet [5]	88.81	82.48	60.83	80.82	78.34	86.06	88.31	90.87	88.77	87.03	68.65	66.90	79.26	79.71	74.67	80.10
GWD [39]	89.66	84.99	59.26	82.19	78.97	84.83	87.70	90.21	86.54	86.85	73.47	67.77	76.92	79.22	74.92	80.23
KLD [43]	89.92	85.13	59.19	81.33	78.82	84.38	87.50	89.80	87.33	87.00	72.57	71.35	77.12	79.34	78.68	80.63
O-RCNN [7]	89.84	85.43	61.09	79.82	79.71	85.35	88.82	90.88	86.68	87.73	72.21	70.80	82.42	78.18	74.11	80.87
KFIoU [44]	89.44	84.41	62.22	82.51	80.10	86.07	88.68	90.90	87.32	88.38	72.80	71.95	78.96	74.95	75.27	80.93
Ours	89.53	85.49	60.75	81.57	79.18	86.06	90.13	90.1	85.18	88.42	69.56	70.32	77.45	79.68	79.79	80.88

and complex background of SP, we design a four-stage frequency feature refinement process, which can effectively extract local, global, and contextual information. Local features fully mine key information, global features enhance the overall understanding of the scene, and contextual features supplement the surrounding environment information, which helps to accurately locate the densely packed targets and improve the detection performance in complex scenes.

3) The mAP of the proposed method on the DOTA-v1.0 dataset is 80.88%, which is only 0.05% lower than the highest method. Compared with other feature extraction methods using only convolution or transformer structure, we analyze the advantages and disadvantages of the two methods, extract local features to focus on key position information, capture global features to model the scene macro structure, extract contextual features to make full use of effective background information, and thus enhance the model object detection performance. It also shows the effectiveness of multiple feature fusion.

The detection effect comparison between the proposed method and other methods on the DOTA-v1.0 dataset is shown in Fig. 7. Different objects are represented by different colors, and their corresponding categories and scores are marked.

From Fig. 7, we can obtain the following results.

1) Column (a) demonstrates that when the object size and direction are varied and the predicted box positioning is difficult, RoI Trans., ReDet, and O-RCNN miss more planes than other methods, while SASM method only detects a small number of SVs. In contrast, the proposed method can detect most objects. The four-stage feature refinement process can effectively extract multiscale features and improve the detection ability of the model for objects of various sizes. At the same time, the sample collaborative optimization loss function can facilitate the connection between classification and regression branches, guiding the accurate positioning of the predicted boxes and improve the detection performance of the model when dealing with targets in different directions.
2) Column (b) demonstrates that in the case of crowded object arrangement, RoI Trans. and ReDet are difficult to frame the object location correctly, while the proposed method can basically classify and locate the object accurately. The frequency feature refinement process is useful for making full use of information and improve the detection performance of crowded targets.
3) Column (c) demonstrates that SASM and ReDet are prone to missing and misdetecting objects in the case of occlusion, while the proposed method can still effectively detect the category and location. The context attention mechanism makes up for the missing information of the occluded object by extracting the features of the surrounding environment, allowing the model to accurately identify and locate the occluded object, thereby improving its detection accuracy.
4) Column (d) demonstrates the detection results of small objects by different rotated object detection methods. The RoI Trans., SASM, ReDet, and O-RCNN have difficulty detecting small tanks, while the RoI Trans., SASM, REDET, and O-RCNN all miss large vehicles. Even in the presence of a complicated background and small target, the proposed model effectively detects the targets, which reflects the effectiveness of the proposed method for the interaction of frequency feature extraction. Furthermore, the position attention calculation of local features aids in the correct localization of small targets by focusing on vital information mixed with global features, thereby eliminating the interference of redundant information.
5) Columns (e) and (f) illustrate the model's performance on blurred images and small object detection tasks, respectively. As shown in the results, although the proposed model demonstrates competitive overall detection performance, it still faces certain challenges when dealing with blurred images and extremely small objects. Specifically, it tends to exhibit noticeable instances of missed detections and false positives, with some small objects failing to be accurately identified. Compared to other state-of-the-art methods, the model shows room for improvement in these aspects. This limitation may stem from constraints in the model's feature extraction and representation capabilities, particularly in capturing fine-

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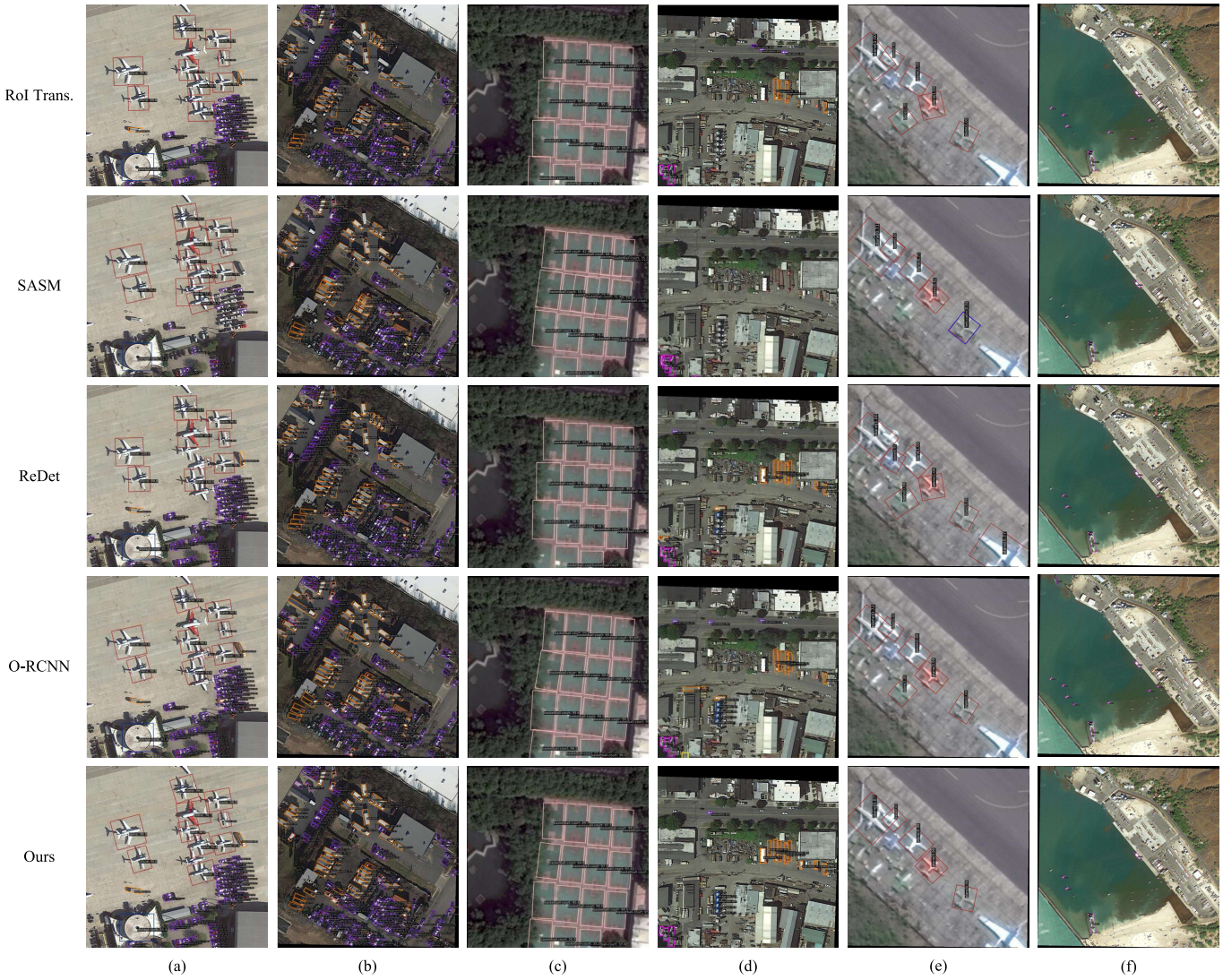


Fig. 7. Comparison of visualized detection results of different methods on the DOTA-1.0 dataset. (a) Depicts a scene with multiple objects that vary in size and direction. (b) Shows a scene with crowded object arrangement. (c) Shows a scene where objects are in an occluded environment. (d) Displays a scene with a mix of various types of small targets. (e) Presents a scene with blurry objects, while (f) displays a scene with small objects.

grained details and distinguishing targets from complex background noise. Therefore, future work should focus on further optimizing the model architecture and training strategies to enhance its robustness and accuracy in detecting blurred and small-scale targets.

D. Ablation Studies

To verify the effectiveness of each module in the proposed method, the following ablation experiments are conducted: the rationality and validity of channel division of the frequency splitter, the necessity of the FIM, the window size and iteration times of the CTB, as well as the sample collaborative optimization loss function. All the ablation experiments were performed on the DOTA-v1.0 dataset.

1) Frequency Splitter: In order to verify the validity of the frequency splitter module, feature channels are divided according to different proportions and then tested as different input features. The experimental results are shown in Table III, where C represents the number of channels and $[0,(1/2)C]$ represents the first half of the channel in the feature map.

TABLE III
ABLATION STUDY RESULTS OF DIFFERENT CHANNEL SPLITTING RATIOS IN THE FREQUENCY SPLITTER

local	global	contextual	mAP
$[0,\frac{1}{2}C]$	$[\frac{1}{2}C,C]$	$[0,\frac{1}{2}C]$	76.45
$[0,\frac{1}{2}C]$	$[\frac{1}{2}C,C]$	$[\frac{1}{2}C,C]$	76.51
$[0,\frac{1}{2}C]$	$[\frac{1}{2}C,C]$	$[0,C]$	**76.87**
$[\frac{1}{2}C,C]$	$[0,\frac{1}{2}C]$	$[0,\frac{1}{2}C]$	74.95
$[\frac{1}{2}C,C]$	$[0,\frac{1}{2}C]$	$[\frac{1}{2}C,C]$	75.32
$[\frac{1}{2}C,C]$	$[0,\frac{1}{2}C]$	$[0,C]$	73.94

From Table III, we can obtain the following results.

a) The first three lines of experiments indicate that high-frequency features are used to extract local information and low-frequency features to extract global information.

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TABLE IV
ABLATION STUDY RESULTS OF THE FIM

Path	PCB	STB	CTB	mAP
Single	✓			73.07
		✓		74.45
			✓	74.11
Dual	✓	✓		75.92
	✓		✓	75.48
		✓	✓	76.26
Triple	✓	✓	✓	**76.87**

TABLE V
ABLATION STUDY RESULTS ON THE EFFECTIVENESS OF PCB, STB, AND CTB

CNN-PCB	PCB	MHSA-STB	STB	MHSA-CTB	CTB	mAP
✓		✓		✓		74.78
✓			✓	✓		76.02
✓		✓			✓	75.57
✓			✓		✓	76.69
	✓	✓		✓		75.06
	✓		✓	✓		76.35
	✓	✓			✓	75.78
	✓		✓		✓	**76.87**

The last three lines used low-frequency features to extract local information and high-frequency features to extract global information. When the channel division of context features is consistent, the performance of extracting local information using high-frequency features is better than that using low-frequency features, and the detection performance is improved by 1.5%, 1.19%, and 2.93%, respectively. It can be seen that high-frequency features are suitable for local feature extraction, while low-frequency features are more suitable for global feature extraction.

b) The experimental results of the first three rows in the table show that the detection effect of using high frequency, low frequency, and full feature to extract context information is 76.45%, 76.51%, and 76.87%, respectively. Therefore, the full feature performs best in constructing context features, which can capture the surrounding environment information more comprehensively and improve the detection performance of the model.

2) Feature Interaction Module: The FIM consists of three branches: PCB, STB, and CTB. This study presents ablation tests of single, dual, and triple branches to assess the efficacy of each branch in the module. When a single branch is employed, the high- and low-frequency features are not separated, and the experimental results are displayed in Table IV.

From Table IV, we can obtain the following results.

a) Using only PCB, STB and CTB modules, the results show that the mAP reaches 73.07%, 75.45%, and 75.11%, respectively. It can be seen that the extraction of global features can improve the detection effect best.

b) The experiment of dual-path branch is PCB + STB, PCB + CTB, STB + CTB three combination methods, and the mAP reaches 75.92%, 75.48%, and 76.26%, respectively. Compared with PCB alone, the addition of STB and CTB improved the detection performance by 2.85% and 2.41%, respectively, indicating that only local features extraction is insufficient, which affects the accuracy of detection. The detection performance of the structure with STB is improved by 0.44% compared with the structure with CTB, because the former is more comprehensive and complete in modeling global information, which can more effectively capture the long-distance dependencies in the image. In addition, the STB + CTB combination of transformer structure can improve the detection performance by 3.19%, 1.81%, and 2.15% compared with the single-path branch structure. This is because the single-path branch only focuses on local information or global information, while the STB + CTB combination can complement each other and enhance the overall performance of the model.

c) The triple-path branch is composed of PCB + STB + CTB, and the mAP reaches 76.87%. Compared to the dual-path branch, the detection performance improves by 1.25%, 1.39%, and 0.61%. Compared to the single-path branch, the detection performance improves by 3.8%, 2.42%, and 2.76%. The experimental results further demonstrate the complementary effectiveness of local features, global features, and contextual features. The interaction of the three features can make up for their respective shortcomings, provide more comprehensive information for the model, and significantly improve the target detection performance of the model in complex scenes.

In order to further verify the validity of each submodule design, the basic convolution and multihead attention mechanism are used to replace the corresponding module, while other structures remain unchanged, where CNN-PCB means using CNN instead of PCB and MHSA-STB and MHSA-CTB mean using MHSA instead of STB and CTB, respectively. The experimental results are shown in Table V.

From Table V, we can obtain the following results.

a) As utilizing PCB in structures with the same global feature branch and context feature branch, the mAP increases by 0.28%, 0.33%, 0.21%, and 0.18%, respectively, as compared to the structure of convolution. It is clear that the use of PCB in the model may effectively capture local information by leveraging position attention, hence improving the detection performance.

b) In structures with the same local feature branches and contextual feature branches, the mAP of the model using STB improves by 1.24%, 1.12%, 1.29%, and 1.09%,

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TABLE VI
ABLATION STUDY RESULTS OF WINDOW SIZE AND ITERATION

window size	iteration	mAP
(8,1,1,1)	1	76.11
(8,4,1,1)	1	**76.87**
	2	76.78
(8,4,2,1)	1	76.45
	2	76.37

respectively, compared to the base MHSA module. It can be seen that STB enhances the long-distance modeling ability of the model, and through the self-focusing attention mechanism, it can pay more attention to the key parts and reduce the interference of redundant information.

c) In the structure with the same local feature branch and global feature branch, the model using the CTB outperforms the basic MHSA model by 0.79%, 0.67%, 0.72%, and 0.52% in terms of mAP, respectively. This indicates that the CTB effectively utilizes contextual features and fully exploits surrounding environmental information, thereby further enhancing the model's detection performance.

d) Using the combination structure of CNN and MHSA module, mAP reaches 74.78%. PCB + STB + CTB combined structure, mAP reaches 76.87%, an increase of 2.09%. The experimental results show that the proposed method can not only effectively extract local features but also capture global information and significantly improve the detection performance of the model by supplementing the context information.

3) Contextual Transformer Block: The contextual transformer block divides the hyperpixel tokens by window size, updating the hyperpixel features through continuous iteration. In this article, we conduct ablation experiments on the window size and iteration times of the four stages, and the experimental results are shown in Table VI.

From Table VI, we can obtain the following results.

a) When the number of iterations is 1 and the partition window size of the four stages is (8, 1, 1, 1), the mAP value is 76.11%. When the number of iterations remains the same and the window size is increased, the mAP reaches 76.87% and 76.45%, respectively, increasing by 0.76% and 0.34%. This indicates that appropriately increasing the window size can expand the receptive field and enable the super-pixel token to cover a wider range of context information, thus improving the detection effect.

b) When the number of iterations is 1, the mAP of the model with a window size of (8, 4, 2, 1) is reduced by 0.42% compared to the model with a window size of (8, 4, 1, 1). This shows that the application of super-pixel token in the first two stages has a good effect, but in the third stage, it will affect the detection performance, because the super-pixel token is mainly to solve the redundant information of shallow features.

c) When the window size is unchanged, increasing the number of iterations does not significantly improve the performance of the model but increases the complexity of the model. Therefore, the model has the best effect when the number of iterations is 1.

TABLE VII
ABLATION STUDY RESULTS OF THE HYPERPARAMETER

α	mAP
4	76.68
5	76.80
6	**76.87**
7	76.79

TABLE VIII
ABLATION STUDY RESULTS OF THE SAMPLE COLLABORATIVE OPTIMIZATION LOSS

w_{pos}	w_{neg}	mAP
-	-	76.23
✓		76.46
	✓	76.58
✓	✓	**76.87**

4) Sample Collaborative Optimization Loss: The weight term w_{pos} includes a hyperparameter α, and to achieve optimal performance, an ablation study on this hyperparameter is conducted. As shown in Table VII, larger values of α allow high-quality positive samples to receive greater positional weights. It can be observed that when $\alpha = 6$, the model achieves the best detection performance.

In order to verify the effectiveness of the sample collaborative optimization loss designed in this article, an ablation experiment with positive and negative sample weights is designed, and the experimental results are shown in Table VIII.

From Table VIII, we can obtain the following results.

a) When no positive/negative sample weight factor is used, the model's mAP is 76.23%. With the use of only positive sample weights, the mAP increases to 76.46%, a 0.23% improvement, indicating that assigning higher weights to high-quality positive samples enhances detection accuracy. When only negative sample weights are used, the mAP reaches 76.58%, a 0.35% improvement. This suggests that assigning higher weights to negative samples that are far from the object, that is, applying a higher penalty, helps the model quickly learn to distinguish these negative samples, reducing interference during model training.

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b) The mAP value of a model using only positive sample weights is 0.12% lower than that of a model using only negative sample weights, implying that for some low-quality anchor boxes, assigning them only small weights, the model may still pay too much attention to them. On the contrary, low-quality anchor boxes are assigned a higher negative sample weight, allowing them to be detected as negative samples more rapidly, reducing their influence in the loss, reducing model interference, and significantly improving detection accuracy.

c) When both positive and negative sample weights are utilized, the mAP is 76.87%, an increase of 0.41% and 0.29% above the positive and negative sample weights alone. This is because using positive and negative sample weights efficiently balances the model's emphasis on positive and negative samples. Assigning higher weights to positive samples directs the model's attention to learning how to accurately locate and classify objects; assigning higher weights to negative samples, particularly those that are far from the object, can reduce the interference of these samples on model training, prevent the model from focusing too much on the background region, and thus better focus on the object. Furthermore, the addition of positive and negative sample weights causes the classification and regression branches to interact and align the classification and regression scores, hence improving the model's detection effect.

V. Conclusion

We propose a frequency feature refinement vision transformer (FFRViT) for rotated object detection, effectively addressing the challenge of detecting rotated objects in remote sensing images. The method employs positional attention, self-focusing attention, and contextual attention mechanisms to extract frequency features, obtaining local, global, and contextual information to enhance the model's feature extraction capability in complex scenarios. The sample collaborative optimization loss is designed to balance the model's attention to positive and negative samples while fostering mutual influence between the classification and regression branches, thereby improving detection performance. Experimental results demonstrate that the proposed method excels in detecting objects in complex scenes and achieves higher detection accuracy compared to several mainstream rotated object detection methods.

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